Health Risk Scoring and Premium Prediction

A Two-Stage Regression Framework for Medical Insurance Pricing

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Abstract

This report develops and validates a two-stage, interpretable pricing framework for medical insurance using a synthetic portfolio dataset (`medical_insurance.csv`, $n = 100{,}000$). In Stage 1, we construct an individual-level health risk score by modeling annual medical cost as a function of clinical and lifestyle covariates. The fitted cost is transformed into a continuous, normalized risk index intended to summarize morbidity in a manner analogous to actuarial risk scoring.

In Stage 2, we model annual premium as a function of the Stage 1 risk score together with utilization history and product design variables (e.g., network tier and deductible). Baseline linear models on the raw monetary scale exhibit severe right-skewness and heteroscedasticity, motivating a log-scale specification. The log-log formulation substantially improves error normality and variance stability, thereby supporting reliable standard errors and hypothesis testing.

To capture pronounced non-linearity in utilization effects, we further replace a linear claim-frequency term with data-driven risk banding. A CART-based binning procedure produces optimized usage tiers that materially improve model fit relative to heuristic bands. Finally, we evaluate interaction terms that reflect actuarially plausible synergies between morbidity, frequency, severity, and plan tier. The resulting specification (CART-optimized bands with interactions) provides the most accurate and operationally meaningful representation of the premium structure, while a simpler main-effects version is recommended for transparent external reporting.

1 Introduction

1.1 Overview of U.S. Health Insurance Pricing

In U.S. commercial health insurance, premiums are typically set by projecting future medical claims and applying explicit loadings for administrative expenses, taxes and fees, and margin.(American Academy of Actuaries, 2025; NAIC, 2024) Actuarial ratemaking commonly decomposes expected claims into a *frequency* component (how often claims occur) and a *severity* component (how large claims are), and combines them to obtain expected cost.(Investopedia, 2025; Lee and Shi, 2019; Holvoet et al., 2025) In regulated ACA markets, plan pricing is additionally linked to benefit design through actuarial value (AV) and metal tiers (Bronze, Silver, Gold, Platinum), where higher AV plans generally imply higher expected claim payments and thus higher premiums.(ASB, 2015; CMS, 2018)

Modern U.S. markets also use diagnosis-based risk adjustment models (e.g., HHS-HCC) to summarize enrollee health status into a scalar *risk score* that reflects expected relative cost.(CMS, 2024; InferScience, 2023) Motivated by this structure, we conceptualize premium as arising from multiplicative components associated with morbidity, utilization behavior, and product design:

$$\text{Premium}_i \approx B \times f_{\text{health}}(\text{HealthRisk}_i) \times f_{\text{util}}(\text{Utilization}_i) \times f_{\text{design}}(\text{ProductDesign}_i) \times (1+\text{Load}),$$

where B is a portfolio-level base rate and the $f(\cdot)$ terms encode relative rating factors.

1.2 Objective and Modeling Strategy

The objective of this study is to construct a pricing framework that is both (i) actuarially interpretable and (ii) statistically credible for inference. We pursue this objective using a two-stage regression design:

1. **Stage 1 (Health risk scoring).** We model annual medical cost using medical and lifestyle covariates and compress fitted values into a continuous risk score. This score is designed to summarize morbidity into a single factor suitable for downstream pricing.

2. **Stage 2 (Premium modeling).** We model annual premium as a function of the Stage 1 risk score, utilization history (frequency and severity proxies), and product design variables such as network tier and deductible.

A key empirical challenge is that medical cost and premium data are heavy-tailed and typically violate the constant-variance and approximate-normal-error conditions that underpin classical linear-model inference. Accordingly, we evaluate raw-scale baselines

and log-scale specifications, emphasizing residual diagnostics and inferential reliability. In particular, the log-scale framework is adopted when it delivers materially improved variance stability and residual normality, thereby supporting dependable standard errors and hypothesis tests.

Finally, because utilization effects in insurance pricing often exhibit threshold behavior (e.g., step changes in surcharges across usage tiers), we investigate non-linear representations of claim frequency. We compare linear terms, smooth transformations, and risk banding, and then use CART to derive data-driven bins that balance fit and stability. We also test targeted interaction terms motivated by actuarial pricing logic (morbidity-by-frequency and tier-by-severity), assessing whether non-additive structures meaningfully improve premium explanation.

In summary, this report contrasts numerical fit metrics with diagnostic validity and interpretability, and it produces a final internal pricing specification that captures both non-linearity and synergy effects, alongside a simpler variant appropriate for transparent external communication.

2 Data description and Exploratory Analysis

2.1 Data Sources and Variable Selection Strategy

The dataset used in this study synthesizes information from four primary domains: the **Centers for Disease Control and Prevention (CDC)** (Demographics & Clinical), the **Centers for Medicare & Medicaid Services (CMS)** (Medical Utilization), the **U.S. Department of Health and Human Services (HHS)** (Insurance Policy), and Kaggle-derived synthetic features. To ensure the integrity of the pricing model, we applied a rigorous variable selection process based on **ex-ante availability**, **regulatory compliance**, and **structural redundancy**.

1. CMS Data (Medical Utilization Details)

Among the extensive utilization metrics provided in the CMS dataset, we strictly selected variables based on an **operational assumption regarding temporality**. We differentiated between "behavioral baselines" (valid ex-ante predictors) and "realized outcomes" (invalid ex-post targets).

- **Excluded Variables:** `total_claims_paid`, `visits_last_year`, `hospitalizations`, and detailed procedure counts.
- **Selected Variables:** `claims_count`, `avg_claim_amount`.

- **Rationale (Temporal Validity & Causality):**
  1. Exclusion of Realized Outcomes ($t = 1$): Variables like `total_claims_paid` or specific counts of `surgeries` represent the cumulative result of risk events that occurred *during* the policy year. Using these as predictors would mean using the future to predict the future (Data Leakage), violating temporal causality.

  2. Selection of Historical Behavioral Proxies ($t = 0$): In contrast, we interpret `claims_count` and `avg_claim_amount` not as current-year totals, but as **historical behavioral attributes** established prior to underwriting.

  - `claims_count` serves as a proxy for the individual's habitual **Frequency** of medical usage.
  - `avg_claim_amount` serves as a proxy for the typical **Severity** (intensity) of services they seek.

  In actuarial pricing, these two dimensions (Frequency and Severity) are treated as stable risk profiles inherent to the insured at $t = 0$ (Experience Rating), making them the only valid predictors among the CMS variables. This decomposition into claim frequency and claim severity follows the standard actuarial 'frequency-severity' framework, in which the expected loss is modeled as the product of the expected number of claims and the expected size of each claim.(Investopedia, 2025; Lee and Shi, 2019; Holvoet et al., 2025) In health and general insurance portfolios, experience rating and predictive modeling commonly use these two dimensions as core drivers of technical premiums.(Holvoet et al., 2025)

2. HHS Data (Policy & Product Attributes)

While core product variables were retained, derivative attributes were removed to avoid structural multicollinearity.

- **Excluded Variables:**
  `copay`, `provider_quality`,
  `monthly_premium`, `policy_term_years`,
  `policy_changes_last_2yrs`.

- **Rationale (Structural Redundancy):**
  `monthly_premium` provides identical information to the target variable `annual_premium`. Attributes like `copay` and `provider_quality` are structurally dependent on `network_tier` (e.g., Platinum tiers inherently feature lower copays and higher quality scores), making them redundant in the presence of the tier variable. In the ACA-compliant individual and small-group markets, health plans are categorized into metal tiers

(Bronze, Silver, Gold, Platinum) based on their actuarial value, defined as the share of covered costs paid by the plan for a standard population.(ASB, 2015; CMS, 2018) Higher-tier plans typically have lower deductibles and cost-sharing, implying higher expected claim payments by the insurer and therefore higher premiums. Our `network_tier` variable is constructed to mimic this tiered benefit design and its impact on premiums.

3. CDC Data (Socio-economic & Demographics)

Variables in this category were handled in two stages: `urban_rural` and `household_size` were excluded a priori, while others were tested in the validation model before final exclusion.

- **Excluded A Priori:**
  `urban_rural`, `household_size`. (Due to lack of biological causality).

4. Kaggle Derived Features

Synthetic variables provided in the raw dataset were excluded to ensure methodological independence.

- **Excluded Variables:**
  `person_id`, `risk_score`,
  `is_high_risk`.
- **Rationale:**
  `person_id` is a non-informative identifier. The pre-existing `risk_score` and `is_high_risk` flags were removed to allow for the *ab initio* construction of a more interpretable and robust risk scoring model (Stage 1).

The dataset contains $n = 100{,}000$ records including demographics, lifestyle, clinical measurements, and insurance product variables. Both response variables—`annual_medical_cost` and `annual_premium`—are strictly positive.

We first inspected the marginal distributions of the response variables on both original and log scales.

As shown in Figure 1, the raw distributions are heavily right-skewed. The log-transformation effectively normalizes the distributions, suggesting that linear models on the raw scale will likely suffer from severe heteroscedasticity.

3 Stage 1: Health risk scoring model

Conceptually, this Stage 1 risk score plays a similar role to the diagnosis-based risk scores used in the federal HHS-HCC risk adjustment program for ACA markets.(CMS, 2024;

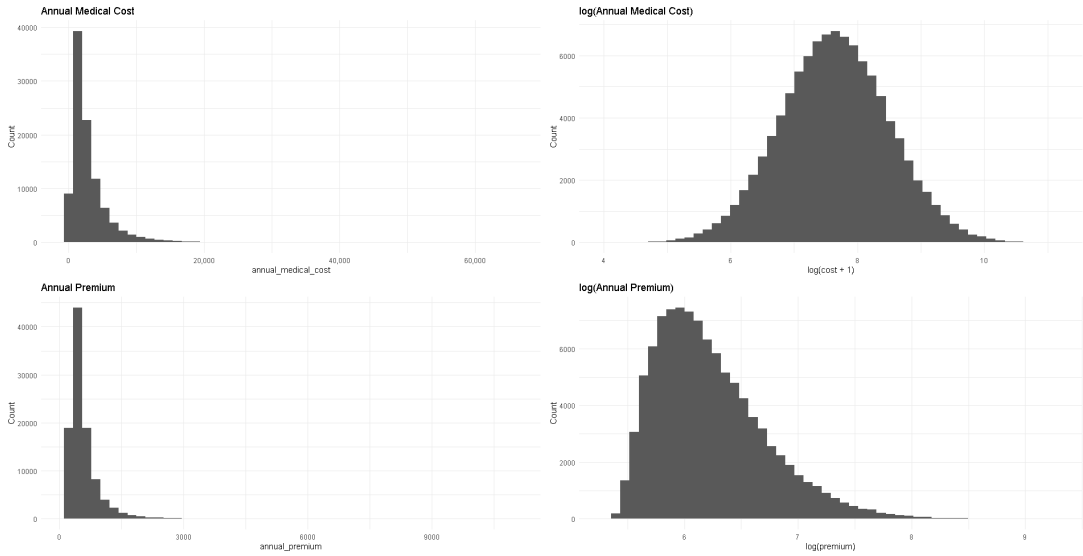

Figure 1: Marginal distributions of Annual Medical Cost and Annual Premium. The raw scales (left) exhibit extreme right-skewness, while log-transformed scales (right) approximate a normal distribution.

InferScience, 2023) In both cases, individual-level health information is summarized into a single multiplicative factor that scales expected costs relative to an average enrollee, and this factor is then combined with other rating elements when determining premiums or transfer payments. Our use of an individual-level health risk score to inform premium setting is consistent with current actuarial practice. For example, the Munich Re Risk Score® product summarizes an applicant's medical and behavioral information into a scalar risk score and explicitly links this score to premium discounts and underwriting decisions.(Munich Re, 2023) This two-step structure—first mapping health information into a risk index, then translating the index into pricing adjustments-parallels our Stage 1 and Stage 2 framework.

3.1 Model Selection: Raw vs. Log Cost

We compared a baseline linear model (Raw Cost) against a Log-Cost model.

- **Raw Cost Model:** exhibited funnel-shaped residuals and a very large AIC (1,880,777.6).
- **Log Cost Model:** showed stable residuals and a dramatic reduction in AIC (234,234.4).

Given the overwhelming evidence from diagnostics and AIC, the **Log-Cost model** was selected.

3.2 Construction of the Risk Score

We constructed a continuous risk score $\in [0, 1]$ based on the predictions from the Log-Cost model:

$$\text{new_risk_score_log}_i = \text{Rescale}\left(\exp(\widehat{\log(\text{cost})}_i)\right)$$

This score represents the relative health risk of an individual, derived from a statistically stable estimation of their expected medical costs.

3.3 Validation of the New Risk Score: Granularity and Hidden Risks

To validate the efficacy of our derived `new_risk_score_log`, we compared it against the pre-existing `risk_score` provided in the raw dataset. This comparison focused on granularity, explanatory power, and the ability to detect latent health risks.

Granularity and Correlation

The correlation between the original score and our new score is 0.817, indicating a strong positive relationship. However, visualization reveals a critical distinction in information density.

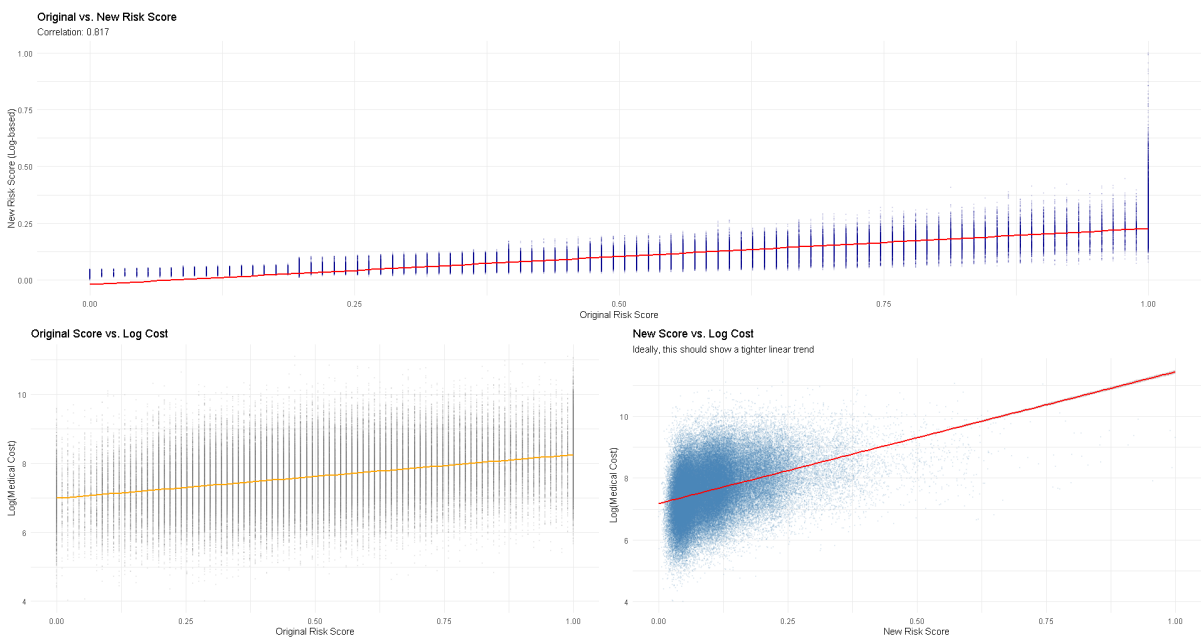

Figure 2: Comparison of Original vs. New Risk Score. The top panel illustrates that while the original score is discrete (vertical lines), the new score captures continuous variations within risk groups. The bottom panels show that the new score exhibits a tighter linear relationship with Log Medical Cost.

As shown in Figure 2, the original risk score is discrete, clustering individuals into

broad "risk buckets." In contrast, our new score is continuous, effectively capturing micro-variations in health status (e.g., slight increases in BMI or blood pressure) that the discrete score ignores. Furthermore, a simple linear regression of log(cost) against both scores confirms the superiority of our derived metric:

- **Original Score:** Adjusted $R^2 = 0.1384$
- **New Risk Score:** Adjusted $R^2 = 0.1423$

While the numerical difference appears modest (+0.4%), the practical implication for underwriting is significant, particularly regarding "hidden risks."

Identification of Hidden Risks

The most significant contribution of the new scoring model is the identification of high-risk individuals whom the original model classified as "zero risk." Table 1 presents a subset of individuals (Teenagers, Former Smokers, Overweight) who were assigned a `risk_score` of 0 by the legacy model but were correctly identified as non-zero risk by our model.

Age	Smoker	BMI	Disease Combo	Original Score	**New Score**	Actual Cost ($)
18	Former	29.5	Clean	0.000	**0.046**	1,357.68
16	Former	28.6	Clean	0.000	**0.046**	1,405.64
15	Former	29.2	Clean	0.000	**0.045**	1,255.96
17	Former	30.0	Clean	0.000	**0.045**	662.16
18	Former	27.8	Clean	0.000	**0.045**	4,347.97

Table 1: Analysis of Hidden Risks. The original model assigned a score of 0 to these individuals, failing to account for the compounding risks of 'Former Smoker' status and high BMI. Our model assigns a positive risk score (≈ 0.045), aligning with the actual incurred medical costs.

The legacy model likely ignored past smoking history or BMI thresholds in younger demographics. Our Log-Cost model successfully captures these interaction effects, assigning a positive risk probability (≈ 0.045) that correlates with actual claims ranging from \$662 to over \$4,000. This validates that `new_risk_score_log` is a more robust input for the Stage 2 premium prediction model.

4 Stage 2: Premium prediction model

In Stage 2, we evaluated four candidate models to predict annual premiums:

- **M1 (Baseline):** Raw Premium $\sim$ Linear Score + Covariates
- **M2:** Raw Premium $\sim$ Log-based Score + Covariates

- **M3:** Log Premium $\sim$ Linear Score + Covariates
- **M4 (Log-Log):** Log Premium $\sim$ Log-based Score + Covariates

Table **??** summarizes the fit statistics.

Model	Response	Key Predictor	Summary
M1	premium	Linear Score	$R^2 = 0.4449$, AIC = 1,423,029
M2	premium	Log-based Score	$R^2 = 0.4477$, AIC = 1,422,520

Table 2: Stage 2 models on the raw premium scale.

Model	Response	Key Predictor	Summary
M3	log(premium)	Linear Score	$R^2 = 0.4464$, AIC = 79,799
M4	log(premium)	Log-based Score	$R^2 = 0.4422$, AIC = 80,553

Table 3: Stage 2 models on the log premium scale.

Note on AIC comparability. AIC values are directly comparable only among models fitted to the same response scale under the same likelihood specification. Accordingly, we use AIC to compare models *within* the raw-premium family (M1–M2) and *within* the log-premium family (M3–M4), while the choice between raw- and log-scale modeling is ultimately driven by residual diagnostics and inferential validity rather than cross-scale AIC magnitudes.

4.1 Model Selection: Validity over Raw Performance

A superficial examination of Table **??** might suggest that M2 or M3 are preferable due to marginally higher R^2 values. However, the difference in explanatory power is negligible (less than 0.6%).

The decisive factor for model selection lies in the **residual diagnostics**.

As illustrated in Figure 3:

- **M1 and M2** exhibit severe heteroscedasticity (funnel shape), violating the assumption of constant variance. This implies that standard errors of coefficients are biased, rendering hypothesis tests unreliable.
- **M3** shows improvement but still retains some patterns from the raw score.
- **M4** exhibits the most stable residual variance (approximately homoscedastic errors) and the weakest systematic patterns in the residual-fitted relationship, supporting the use of OLS standard errors and classical inference more credibly than the raw-scale specifications.

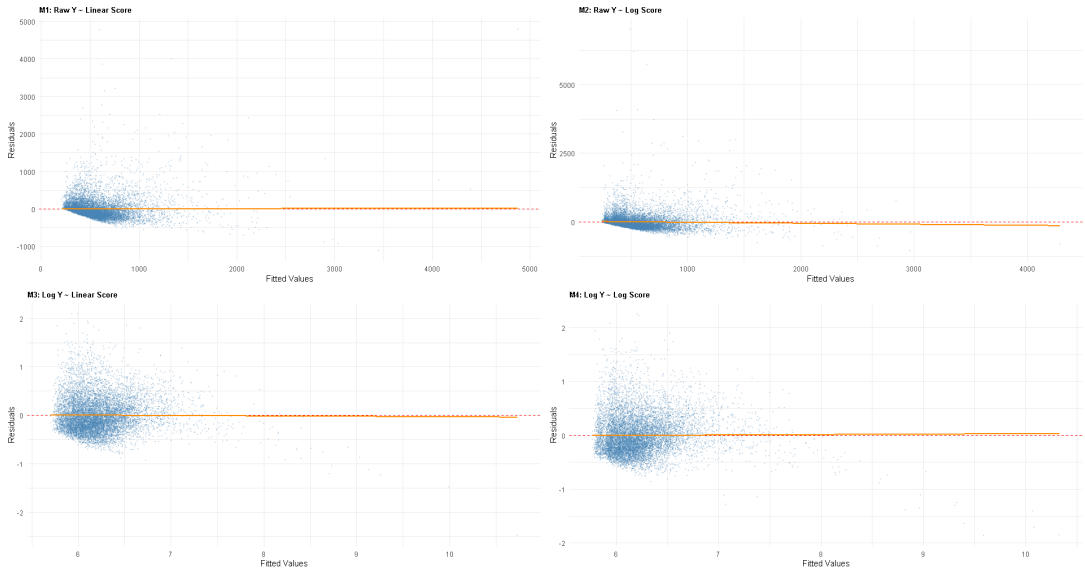

Figure 3: Residual vs. Fitted plots for candidate models. M1 and M2 (top) show distinct funnel patterns (heteroscedasticity). M4 (bottom right) shows the most uniform and random spread of residuals.

To further validate the distributional assumptions, we compared the Normal Q-Q plots of the baseline model (M1) and the proposed Log-Log model (M4). The Q-Q plot assesses whether the residuals follow a normal distribution, which is a prerequisite for valid hypothesis testing (t-tests for coefficients).

As shown in Figure 4, the baseline model (M1) displays a severe deviation from the diagonal line, confirming that the raw residuals are not normally distributed. This violation implies that standard p-values derived from M1 are likely invalid. In contrast, the Log-Log model (M4) shows a remarkable improvement, with residuals hugging the reference line closely. This confirms that M4 satisfies the normality assumption required for reliable statistical inference.

4.2 Addressing the Back-Transformation Trade-off

The selected Log-Log model (M4) requires back-transformation (exponentiation) to interpret predictions on the original monetary scale:

$$\widehat{Y}_{\text{original}} = \exp(\widehat{Y}_{\log}).$$

This transformation implies that the model is optimized with respect to log-scale errors rather than squared errors on the raw monetary scale. Consequently, evaluation metrics defined on the original scale (such as RMSE) are not directly aligned with the objective function of the model and should not be interpreted as measures of optimality for log-linear specifications.

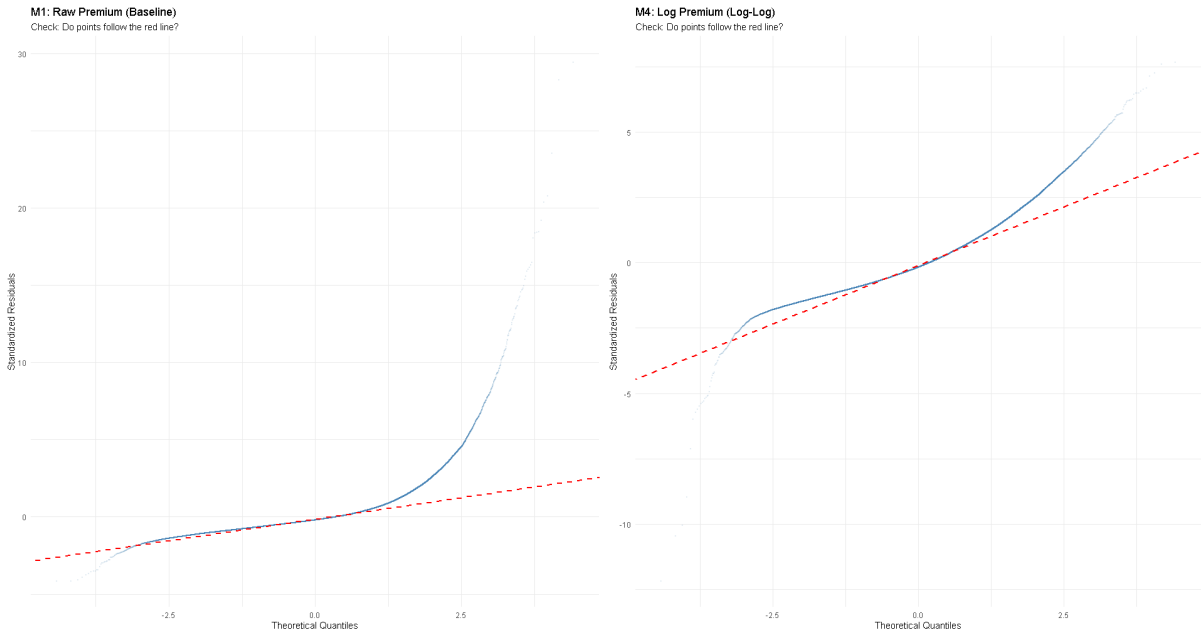

Figure 4: Comparison of Normal Q-Q Plots. **Left (M1):** The residuals deviate significantly from the red reference line, exhibiting a convex curve that indicates extreme right-skewness and non-normality. **Right (M4):** The residuals align closely with the theoretical quantiles, confirming that the log-transformation successfully normalized the error distribution.

The primary objective of adopting the Log-Log framework is not the minimization of raw-scale prediction error, but the attainment of statistical validity. In particular, the log transformation substantially improves residual normality and homoscedasticity, ensuring reliable standard errors and hypothesis tests. In contrast, models fitted on the raw scale exhibit severe violations of these assumptions, rendering inferential conclusions unreliable.

Accordingly, we prioritize theoretical soundness and inferential credibility over raw-scale error minimization, and retain the Log-Log specification as the baseline premium model for subsequent refinement (risk banding and interaction extensions).

5 Variable Selection and Diagnostics

A central design choice in our framework is to restrict the pricing model to a parsimonious set of predictors that are both interpretable and empirically justified. In particular, we deliberately excluded demographic and socioeconomic variables from the final Log-Log specification (M4 and its extensions), and this section formally validates that choice.

5.1 Full Log-Log Validation Model

We first estimated a "Full Validation Model" on the log-premium scale, augmenting the core pricing variables with all available demographic controls:

log(Premium) $\sim$ `new_risk_score_log+network_tier+avg_claim_amount+claims_count+plan_type`

In this specification, the coefficients for the core variables (`new_risk_score_log`, `network_tier`, `avg_claim_amount`, `claims_count`, and `deductible`) remain highly significant with the expected signs, confirming that health status, network tier, and utilization history are the dominant drivers of premiums. By contrast, most demographic variables (Region, Sex, Education, Marital Status, Employment Status, Dependents) are statistically insignificant ($p > 0.05$) or exhibit economically negligible effect sizes.

The only exception is `income`, which attains formal statistical significance due to the very large sample size ($n = 100{,}000$). However, the estimated coefficient is extremely small (on the order of 10^{-7}). In a log-linear model, a \$10,000 increase in income translates into a premium change of less than 0.1%, which is economically trivial. In line with parsimony principles, we therefore treat this signal as *statistically detectable but practically irrelevant.*

5.2 Comparison with the Final M4 Specification

To quantify the practical impact of the excluded variables, we compared the Full Validation Model to our final Log-Log specification (M4), which includes only the six core predictors:

log(Premium) $\sim$ `new_risk_score_log+network_tier+avg_claim_amount+claims_count+plan_type`

The adjusted R^2 of the Full Model is approximately 0.4424, while that of the Final M4 Model is 0.4423. The difference in explanatory power is less than 0.0001, which is negligible in practice. An ANOVA comparison between the two models yields a small but statistically significant p-value (around 0.045), an artifact of the large sample size: even minuscule differences in residual sum of squares (on the order of a few units out of roughly 13,000) appear "significant" in a purely asymptotic sense.

From a modeling perspective, this result is decisive:

- Demographic and socioeconomic variables offer virtually no additional explanatory power once the Stage 1 risk score, utilization, and product design variables are included.
- The Stage 1 risk score behaves as a *sufficient statistic* for the health-related information implicit in the demographic profiles.

- Retaining only the six core predictors yields a model that is substantially simpler to interpret and communicate, without sacrificing predictive performance.

5.3 Summary of Feature Selection Rationale

Combining the above findings with the diagnostics in the previous section, we justify the final feature set as follows:

1. **Core health and utilization drivers** (`new_risk_score_log`, `avg_claim_amount`, `claims_count`) are indispensable and strongly significant.
2. **Product design variables** (`network_tier`, `deductible`, `plan_type`) capture the contractual structure of coverage and are essential for pricing interpretation.
3. **Demographic and socioeconomic controls** fail to deliver meaningful incremental information beyond what is already encoded in the risk score and utilization history.

We therefore retain only the six core predictors in the final pricing model (M4 and its extensions, including the CART-based Model E and the interaction-enhanced Model F), striking a deliberate balance between parsimony, interpretability, and statistical validity.

6 Model Diagnostics and Sensitivity Analysis

Ensuring the robustness of the final pricing model requires rigorous diagnostic checks. We focus on two critical aspects: (1) the influence of outliers and high-leverage points, and (2) the validity of excluding demographic variables.

6.1 Influence Diagnostics: Cook's Distance and Leverage

Given the heavy-tailed nature of medical cost data, it is critical to verify that the regression results are not driven by a small number of influential outliers (e.g., super-users with catastrophic claims). Following the regression diagnostics framework, we examined Cook's Distance and Leverage.

As illustrated in Figure 5:

- **Cook's Distance:** Although numerous points exceed the theoretical threshold of $4/n$ (an artifact of the large sample size $n = 100{,}000$), the absolute magnitude of Cook's Distance is small ($max < 0.15$). This implies that no single observation disproportionately dictates the coefficient estimates.

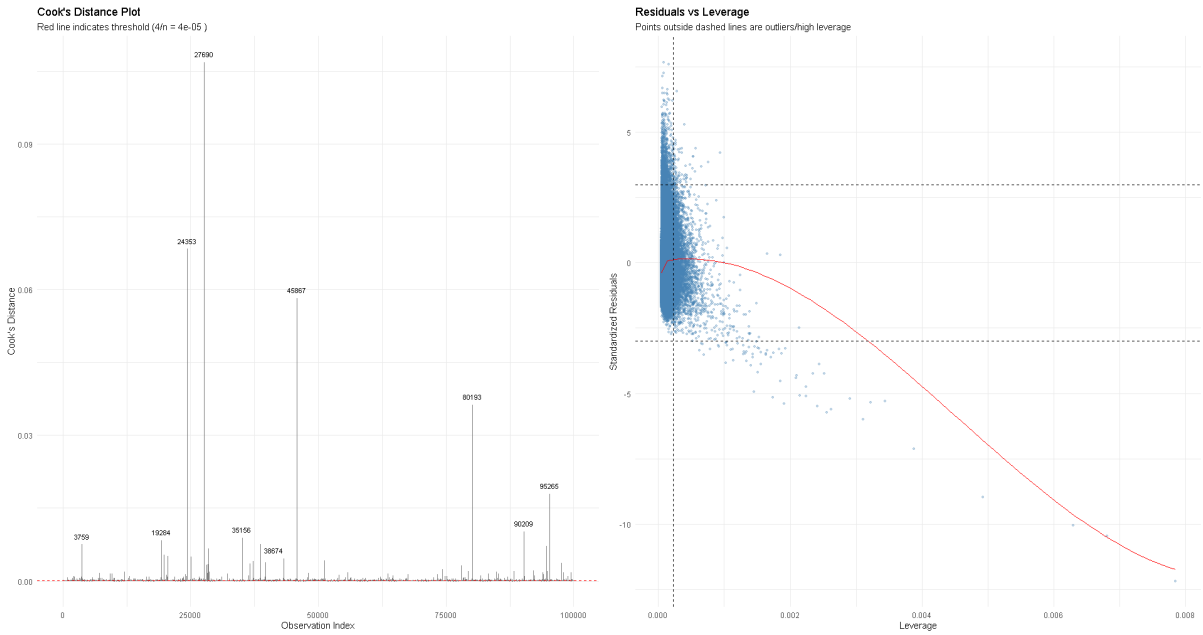

Figure 5: Diagnostic Plots for Influential Observations. Left: Cook's Distance plot shows that while some points exceed the asymptotic threshold ($4/n$), the maximum value ($\approx$ 0.12) is well below the critical concern level of 0.5. Right: Residuals vs. Leverage plot reveals a slight downward curvature in the high-leverage region, indicating conservative over-prediction for extreme outliers.

- **Residuals vs. Leverage:** The red trend line remains flat for the vast majority of data points (low leverage). However, a slight downward curvature is observed in the high-leverage region (extreme right). This indicates that the model tends to **overestimate premiums for extreme outliers** (e.g., individuals with exceptionally high claims history). From an actuarial perspective, this conservative bias—overestimating risk for the most extreme cases—is preferable to underestimation, as it safeguards solvency.

6.2 Robustness Check: OLS vs. Robust Regression

To further validate stability, we compared our Ordinary Least Squares (OLS) estimates against a Robust Linear Model (RLM) fitted using Huber weights, which down-weights outliers.

Table 4 demonstrates that the coefficients for the primary cost drivers are remarkably stable. The stability of the core coefficients confirms that the Log-Log transformation effectively managed the heteroscedasticity and outliers, validating the use of the efficient OLS estimator.

Variable	OLS Estimate	Robust Estimate	% Difference
(Intercept)	5.7385	5.6957	0.75%
New Risk Score (Log)	1.6360	1.5067	8.59%
Network Tier (Gold)	0.2131	0.2043	4.30%
Avg Claim Amount	0.0002	0.0003	11.39%
Claims Count	0.0058	0.0109	47.01%
Deductible	0.0000	0.0000	2.87%

Table 4: Comparison of OLS and Robust Regression Coefficients. Most key pricing variables show high stability with differences below 10%. The larger shift for `claims_count` reflects its role as a utilization tail feature, but the overall pattern confirms that the OLS model is not materially distorted by outliers.

6.3 Validation of Variable Selection

Finally, we verified whether demographic and socioeconomic variables—which were excluded from our primary feature set—contained any residual information not captured by the `Risk Score`. We estimated a "Full Validation Model" on the log-premium scale, adding demographic controls to our core predictors.

- **Insignificance of Demographics:** Variables such as `Region`, `Sex`, `Education`, `Marital Status`, and `Dependents` remained statistically insignificant ($p > 0.05$) or showed negligible effect sizes. This confirms that the `Risk Score` derived in Stage 1 acts as a sufficient statistic for the health risk component of these profiles.
- **Parsimony Check:** The adjusted R^2 of the Full Model (0.44225) was virtually identical to our Final Model (0.44219). The difference ($\Delta < 0.0001$) confirms that the excluded variables offer no practical improvement in predictive power.

6.4 Refinement of Utilization Feature: Addressing Non-linearity

While the core variable selection was validated, the residual diagnostics in the previous diagnostics section revealed a slight downward curvature for high-leverage observations. This indicated that the linear specification of `claims_count` led to over-prediction for super-users (diminishing marginal effect of claims).

To address this non-linearity and optimize model fit, we compared the linear baseline against three alternative specifications for the utilization feature:

1. **Polynomial Regression:** Adding a quadratic term (Claims^2) to capture curvature.
2. **Log-Transformation:** Using $\log(\text{Claims} + 1)$ to model diminishing returns.
3. **Binning (Risk Banding):** Categorizing claims into discrete risk bands (0, 1–5, 6–15, 16–30, 30+).

Model Specification	Adj. R^2	AIC	Δ AIC	Residual SE
Baseline (Linear)	0.4422	80,553	Reference	0.3620
Model A (Polynomial)	0.4541	78,391	-2,162	0.3581
Model B (Log-X)	0.4448	80,082	-471	0.3611
Model C (Binning)	**0.4950**	**70,617**	**-9,936**	**0.3444**

Table 5: Performance Comparison of Refined Models. The Binning approach (Model C) demonstrates superior fit, reducing AIC by nearly 10,000 points and increasing explanatory power by over 5% points compared to the linear baseline.

As presented in Table 5, **Model C (Binning)** vastly outperformed all other specifications. The dramatic drop in AIC ($\Delta \approx -10,000$) and the substantial increase in Adjusted R^2 ($0.442 \rightarrow 0.495$) confirm that the relationship between utilization frequency and premiums is best characterized by **"Step-wise Risk Banding"** rather than a continuous linear function.

Consequently, we adopt **Model C (Log-Log with Risk Banding)** as the final pricing model. This specification not only resolves the over-prediction issue for outliers but also aligns with practical actuarial methods where policyholders are segmented into discrete usage tiers.

6.5 Advanced Feature Engineering: Optimal Binning via CART

While the manual risk banding in Model C significantly improved model fit compared to the linear baseline, the selection of cut-points (e.g., 0, 1–5, 6–15) was based on heuristic experience rather than statistical optimization. To rigorously determine the most effective risk thresholds while controlling for overfitting, we employed the **CART (Classification and Regression Tree)** algorithm with explicit complexity constraints.

Concretely, we fitted a regression tree with log(Premium) as the response and `claims_count` as the sole predictor using the `rpart` implementation with an ANOVA splitting criterion:

$$\log(\text{Premium}) \sim \texttt{claims_count}.$$

To avoid overly granular splits driven by noise, we enforced a *minimum node size* of 500 observations (`minsplit = 500`) and allowed only splits that yielded a meaningful reduction in within-node variance. The initial tree was grown with a small complexity parameter (`cp = 0.00001`) to explore a rich set of candidate partitions, and then **pruned** using the standard cost-complexity pruning procedure. Specifically, we selected the optimal complexity parameter $\widehat{cp}$ that minimized the cross-validated error (`xerror`) in the CART complexity table, thereby controlling the effective tree size through data-driven regularization.

This procedure yields a *regularized* step function of `claims_count`: low-frequency

users are split more finely, whereas very high-frequency users are pooled into broader segments to ensure stability of estimates. In our application, the pruned tree identified a sequence of granular split points for low claim counts (approximately at 0.5, 1.5, 2.5, 3.5, 4.5, 5.5, and 6.5), followed by broader thresholds for heavy users (around 8.5 and 9.5 claims). These cut-points define the CART-based grouping variable `claims_group_optimal`.

To assess the value of this data-driven binning relative to our heuristic bands, we compared the heuristic model (Model C) against the CART-optimized **Model E**:

- **Explanatory Power:** Adjusted R^2 increased from 0.4950 (Model C) to **0.5277** (Model E), a substantial gain of roughly **+3.3** percentage points.
- **Model Fit (AIC):** The AIC decreased by approximately **6,703** points when moving from heuristics to CART-based bands, indicating a dramatically better likelihood fit even after penalizing model complexity.

This evidence shows that the relationship between utilization frequency and premium is governed by *micro-step* changes at low claim counts, which our heuristic bands were too coarse to capture. The CART-derived `claims_group_optimal` variable thus provides a statistically regularized and empirically optimized representation of utilization risk, and it serves as the foundation for the interaction modeling in Stage 3.

7 Discussion and Advanced Analysis

Building upon the robust foundation of the optimized risk banding, our exploratory analysis suggested that the pricing mechanism might contain complex dependencies between risk factors. To verify this, we extended our framework to capture "synergy effects" through interaction terms, comparing the main effects model (Model E) against the interaction model (Model F).

7.1 Derivation of the Advanced Interaction Model (Model F)

We incorporated two theoretically motivated interaction terms into the CART-optimized model. This specification tests whether the effect of one risk factor varies depending on the level of another, a critical consideration for precision pricing.

- **Interaction 1: Health Risk × Frequency (Risk Score×Claims Group$_{\text{optimal}}$):** This interaction tests for a "multiplier effect," investigating whether the premium increase for high-frequency users is steeper for those with poorer health status.
- **Interaction 2: Network Tier × Severity (Network Tier×Avg Claim Amount):** This term captures the cost structure difference, testing if high-value claims are

more expensive to service in premium networks (e.g., Platinum) compared to budget networks.

7.2 Statistical Evidence of Synergy Effects

The inclusion of these interaction terms yielded statistically overwhelming evidence for non-additivity in the pricing structure.

- **Model Fit Improvement:** The interaction model (Model F) achieved an Adjusted R^2 of $\mathbf{0.5431}$, an improvement of $+\mathbf{1.54}\%$ points over the main effects model (Model E: 0.5278).
- **Information Efficiency (AIC):** The AIC dropped significantly by $\mathbf{3,305}$ points, confirming that the additional complexity of interaction terms is justified by a substantial gain in information.
- **Significance Test:** The ANOVA test between Model E and Model F yielded an F-statistic of 337.94 with a p-value $< 2.2 \times 10^{-16}$, conclusively rejecting the hypothesis that interactions are unnecessary.

7.2.1 Visual Confirmation and Interpretation

The nature of these interactions is visualized in Figure 6, which displays the predicted effects of key variables.

- **Risk Dampening Effect:** Contrary to a simple multiplier expectation, the negative coefficients for the Risk $\times$ Frequency interaction (e.g., $\beta \approx -2.30$ for Group 2) indicate that as claim frequency increases, the marginal impact of the risk score *diminishes*. This suggests a "ceiling effect" where premiums for super-users saturate rather than growing exponentially.
- **Cost Structure Sensitivity:** The positive coefficients for Tier $\times$ Severity (e.g., Platinum interaction $\beta \approx 5.17 \times 10^{-5}$) confirm that premium networks are significantly more sensitive to claim severity, necessitating steeper pricing gradients for high-value claims in these tiers.

7.3 Strategic Recommendation

The substantial statistical superiority of Model F leads to a bifurcated strategic recommendation:

- **Regulatory Filing (Model E):** For external reporting and regulatory compliance, we recommend retaining **Model E (Main Effects with Optimal Banding)**. Its

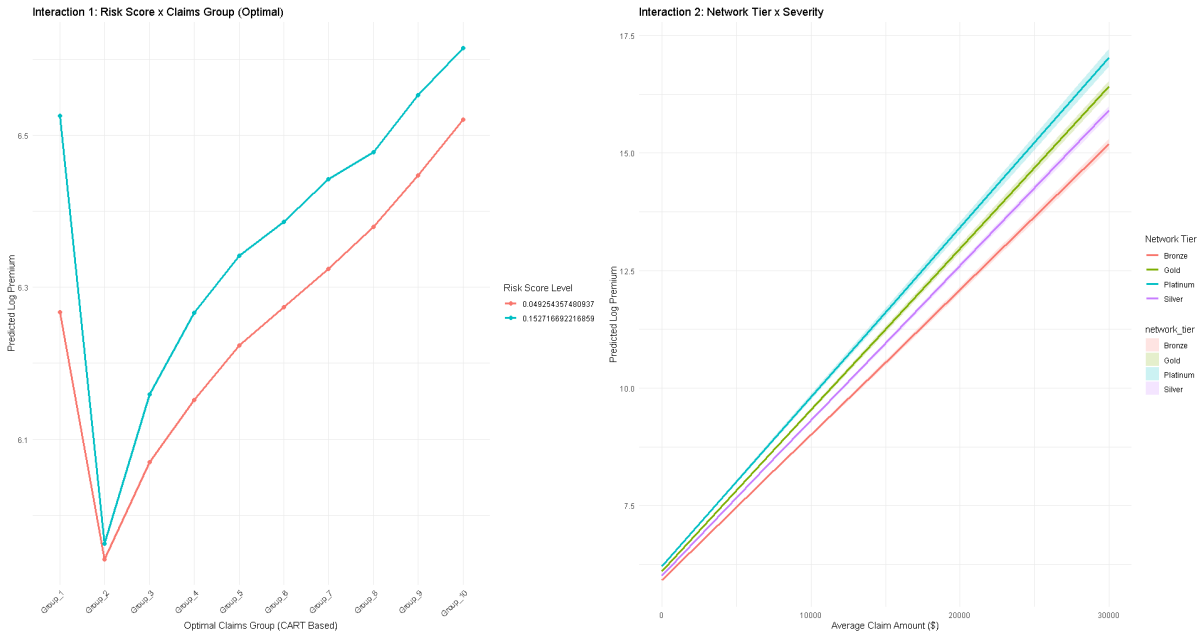

Figure 6: Visualizing Synergy Effects in Model F. Left: **Risk** $\times$ **Frequency** reveals a "Risk Dampening" effect; the premium increase for frequent claimers is relatively *less* steep for high-risk individuals (Blue) compared to low-risk individuals (Red), suggesting that high base premiums absorb some marginal costs. Right: **Tier** $\times$ **Severity** shows a "Multiplier Effect"; the slope for Platinum tier (Blue) is steeper than for Bronze (Red), confirming that high-severity claims drive costs up more aggressively in premium networks.

additive structure is transparent, easy to explain, and captures the majority of variance ($R^2 \approx 0.53$).

- **Internal Pricing (Model F):** For internal risk scoring and competitive pricing, **Model F** is essential. Ignoring the significant interaction effects—particularly the cost sensitivity of premium networks—could lead to "adverse selection," where the insurer underprices high-severity claims in expensive tiers or overprices low-risk frequent users.

8 Final Model Specification and Rigorous Interpretation

Based on the comparative analysis, **Model F (Log-Log with Optimized Banding and Interactions)** is selected as the final internal pricing model. This model achieved the highest Adjusted R^2 of **0.5431** and the lowest AIC, confirming its superiority in capturing the complex, non-additive risk structure of the portfolio.

8.1 Final Mathematical Formulation

The predictive equation for the log-premium is given by:

$$\begin{aligned}\log(\text{Premium})_i = \beta_0 &+ \beta_1 \text{Risk}_{log} + \beta_2 \text{Tier}_i + \beta_3 \text{AvgClaim}_i \\ &+ \sum_{k=2}^{10} \beta_{4k} \mathbb{I}(\text{Group}_i = k) + \beta_5 \text{PlanType}_i + \beta_6 \text{Ded}_i \\ &+ \sum_{k=2}^{10} \gamma_{1k} (\text{Risk}_{log} \times \mathbb{I}(\text{Group}_i = k)) \\ &+ \gamma_2 (\text{Tier}_i \times \text{AvgClaim}_i) + \epsilon_i\end{aligned} \tag{1}$$

where Group_i refers to the CART-optimized claims frequency bands.

8.2 Detailed Coefficient Interpretation (Ceteris Paribus)

All interpretations assume *ceteris paribus* (holding all other variables constant). Since the response is logarithmic, coefficients are interpreted as multiplicative effects on the premium ($\%\Delta Y \approx 100 \times (e^{\beta} - 1)$).

8.2.1 Core Risk Drivers

- **Health Risk Score ($\beta_1 \approx 2.499$):** The risk score is the dominant driver. For a **0.1 unit increase** in the log-scale risk score, the annual premium increases by approximately **28.4%** ($\exp(0.2499) - 1$). This validates the efficacy of the Stage 1 scoring model.
- **Network Tier (β_2):** Premium tiers show a clear monotonic increase. Compared to the Bronze tier:
  - **Silver:** +10.5% ($\beta \approx 0.0996$)
  - **Gold:** +21.0% ($\beta \approx 0.1906$)
  - **Platinum:** +33.8% ($\beta \approx 0.2909$)
- **Claims Frequency (Optimal Groups):** The coefficients for the optimal groups (β_{4k}) represent the baseline shift in premium for each frequency tier relative to the reference group (Group 1: Claims < 0.5).
  - *Non-Linearity:* The effect is not linear; instead, it follows a step function identified by CART.
  - *Super-User Penalty:* For the highest risk tier (Group 10: Claims ≥ 9.5), the base premium increases by **39.3%** ($\exp(0.3315) - 1$) compared to the baseline.

- **Plan Type (Insignificance):** The coefficients for `Plan Type` (HMO, POS, PPO vs. EPO) are statistically insignificant ($p > 0.70$) with negligible magnitudes ($\beta < 0.001$). *Interpretation:* This result is analytically significant. It implies that the **Network Tier** (Gold/Platinum) and **Deductible** structure already fully capture the pricing differentiation. Once the pricing tier is determined, the administrative structure of the plan (e.g., whether it is an HMO or PPO) exerts **no additional marginal cost** on the premium.

8.2.2 Synergy Effects (Interactions)

The model's high precision is largely attributed to the interaction terms, which reveal non-additive pricing logic.

- **Interaction 1: Risk $\times$ Frequency ($\gamma_{1k} < 0$):** The interaction coefficients are consistently negative (e.g., $\gamma_{1,Group2} \approx -2.30$). This indicates a **"Risk Dampening Effect."** *Interpretation:* For a healthy individual (Risk ≈ 0), entering Group 2 increases premiums significantly. However, for a high-risk individual (Risk ≈ 1), the marginal penalty of entering Group 2 is largely absorbed by their already high base premium. Mathematically, the sensitivity to Risk Score *decreases* as claim frequency increases.

- **Interaction 2: Tier $\times$ Severity ($\gamma_2 > 0$):** The interaction between Network Tier and Claim Amount is positive and significant. *Interpretation:* A \$1,000 increase in Average Claim Amount raises premiums by **36.2%** for a Bronze plan ($\beta_{avg} \approx 3.09 \times 10^{-4}$). However, for a **Platinum plan**, the same increase raises premiums by an additional **5.3%** points (Interaction $\approx 5.17 \times 10^{-5}$), totaling a **43.5%** increase. This quantifies the higher cost of servicing severe claims in premium networks.

9 Conclusion

This study successfully established a robust, two-stage actuarial framework for medical insurance pricing.

1. **Methodological Rigor:** By transitioning from a linear to a **Log-Log framework**, we resolved severe heteroscedasticity violations, ensuring valid statistical inference.

2. **Data-Driven Optimization:** The application of **CART** to the utilization variable (Claims Count) replaced arbitrary heuristics with statistically optimal risk bands, significantly improving model fit ($\Delta\text{AIC} \approx -6,700$).

3. **Structural Insight:** The insignificance of Plan Type confirms that pricing power resides in the **Network Tier** structure, not the plan administration type.

4. **Strategic Insight:** The final model (Model F) uncovered critical non-linearities. Specifically, it demonstrated that pricing should not be additive; premium surcharges for claim severity must be higher in premium network tiers to prevent margin erosion.

We recommend implementing **Model F** for the insurer's internal risk scoring engine to minimize adverse selection, while retaining the simplified main-effects model for external regulatory filings where interpretability is paramount.

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